KINGSCLIFF HIGH SCHOOL

ACHIEVEMENT THROUGH ENDEAVOUR

NAME:

TEACHER:

MATHEMATICS

Year 9 5.2

Yearly Exam Practice

Practice Exam 2

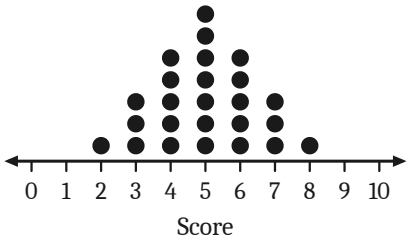
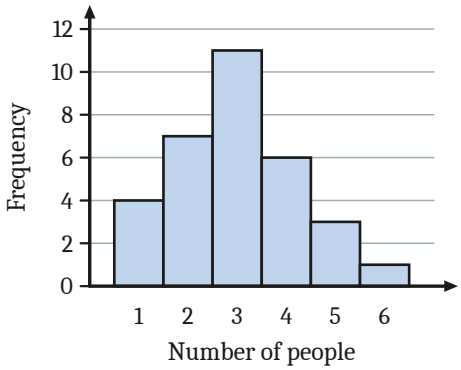
Instructions

1. Working time: 60 minutes
2. Attempt all questions
3. Show ALL working where needed
4. A calculator is permitted
5. Answers are at the back of this booklet

Sections and Marks

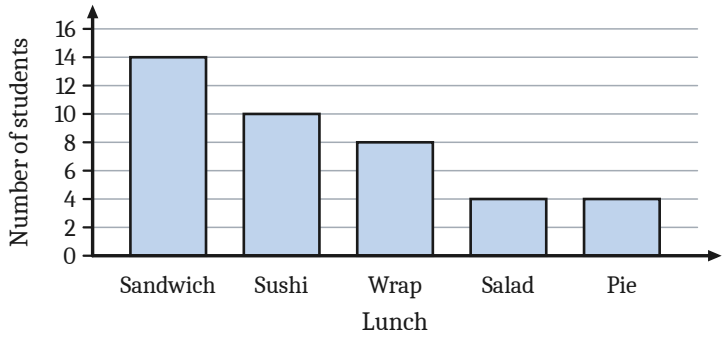
Section 1 - Data	/14
Section 2 - Surface area and volume	/20
Section 3 - Mixed Topics	/30
Total marks	**/64**

Section 1: Data

	Multiple Choice:	Mark
1	Which of the following is **numerical discrete** data? A. The mass of a parcel B. The number of text messages sent in a day C. Hair colour D. The time taken to run 100 m	1
2	The dot plot shows the scores (out of 10) in a times-tables quiz of 25 students. Which statement best describes the data? A. Positively skewed, clustered at 4 to 6 B. Negatively skewed, clustered at 4 to 6 C. Symmetrical, clustered at 4 to 6 D. Symmetrical, clustered at 2 and 8 	1
3	The histogram shows the number of people living in each of 32 houses in a street. What is the mode? A. 3.5 B. 6 C. 3 D. 11 	1
4	These are the marks (out of 70) of 7 students in a maths test: 62, 65, 8, 64, 63, 66, 65 Which measure of centre changes the most if the outlier, 8, is removed? A. The mean B. The mode C. The median D. None of them change	1

Section 1: Data (continued)

Multiple Choice:

		Mark
5	Students were asked what they most often eat for lunch. The column graph shows the results. What percentage of the students chose sandwich?  A. 14% B. 28% C. 35% D. 40%	1

Short Answer:

	Question	Working Out	Mark
6	The stem-and-leaf plot shows the number of minutes that 11 students spent reading on Sunday. Stem | Leaf 1 | 4 9 2 | 2 5 7 7 3 | 1 4 8 4 | 0 2 Key: 2 | 7 = 27 minutes a. Find the mean. b. Find the median. Note: Mean = sum of the scores ÷ number of scores	a. b.	2 1

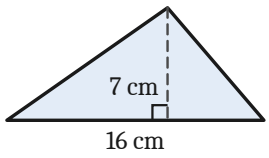
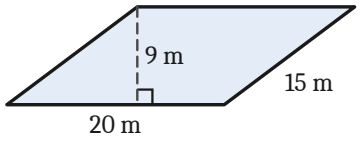
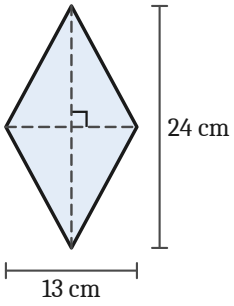
Section 1: Data (continued)

Worded Problems:

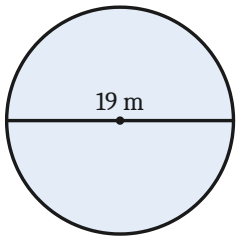
	Question	Working Out	Mark
7	Jordan and Casey recorded how many minutes they spent on homework each night for 10 nights. The results are shown in the back-to-back stem-and-leaf plot. Note: Range = highest score − lowest score	Jordan | Stem | Casey 1 | 2 8 9 7 6 3 | 2 | 1 7 7 5 3 1 1 0 | 3 | 2 4 8 4 | 1 4 9 Key: 3 | 2 = 32 minutes	
a	Find the median number of minutes for each student.		2
b	Which student usually spends more time on homework? Give a reason.		1
c	Which student is more consistent? Find the range of each student to explain.		3
	Total		**/14**

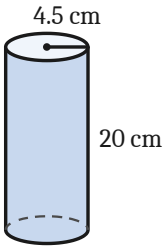
Section 2: Surface area and volume

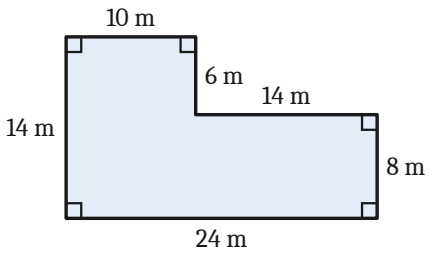
Multiple Choice:

		Mark
1	A tanker carries 2350 L of milk. How many kilolitres is this? A. 23.5 kL B. 235 kL C. 2.35 kL D. 2 350 000 kL	1
2	What is the area of this triangle? A. 112 cm^2 B. 23 cm^2 C. 56 cm^2 D. 11.5 cm^2 	1
3	What is the area of this parallelogram? Note: Area of a parallelogram: $A = bh$ A. 180 m^2 B. 300 m^2 C. 90 m^2 D. 70 m^2 	1
4	What is the area of this rhombus? Note: Area of a rhombus: $A = \frac{1}{2}xy$ (x and y are the diagonals) A. 312 cm^2 B. 78 cm^2 C. 37 cm^2 D. 156 cm^2 	1

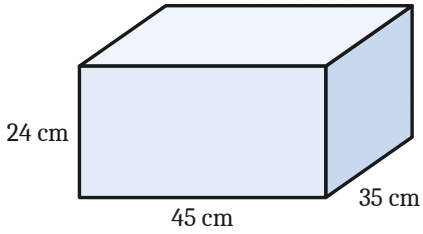
Short Answer:

	Question	Working Out	Mark
5	For this circle, find, correct to 1 decimal place:  a. its circumference b. its area. Note: Circumference: $C = 2\pi r$ Note: Area: $A = \pi r^2$	a. b.	1 1

Section 2: Surface area and volume (continued)			
	Question	**Working Out**	**Mark**
6	A closed poster tube is a cylinder. It has a radius of 4.5 cm and a height of 20 cm.  a. Find its volume, correct to 1 decimal place. b. Find its surface area, correct to 1 decimal place. Note: Volume: $V = \pi r^2 h$ Note: Surface area: $SA = 2\pi r^2 + 2\pi rh$	a. b.	2 2

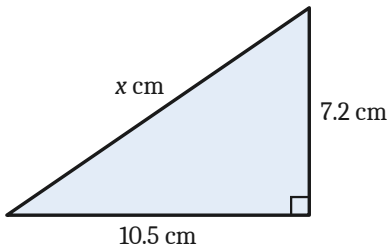
	Section 2: Surface area and volume (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
7	The plan shows an L-shaped timber deck. All of the corners are right angles. Note: Area of a rectangle: $A = lb$ 		
a	A safety rail is fitted around the outside edge of the deck. How many metres of rail are needed?		1
b	Find the area of the deck.		3
c	Decking costs $64.50 per square metre. Find the cost of the decking for the whole deck.		1

Section 2: Surface area and volume (continued)

	Question	Working Out	Mark
8	A closed storage crate is shaped like a box. It is a rectangular prism 45 cm long, 35 cm wide and 24 cm high. 		
a	Find the volume of the box in cm^3. Note: Volume of a rectangular prism: $V = lwh$		1
b	How many litres does the box hold? Note: 1 L = 1000 cm^3		1
c	The outside of the crate is to be covered in plastic film. Find the surface area of the box. Note: Surface area = the total area of the 6 faces; area of a rectangle: $A = lb$		3
	Total		**/20**

Section 3: Mixed Topics (Chapters 1–8)

Multiple Choice:

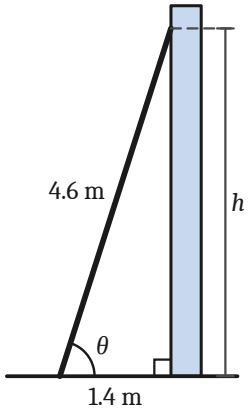
		Mark
1	If $p = 6$ and $q = 2$, evaluate $2p^2 + q$. A. 14 B. 26 C. 74 D. 146	1
2	Find the value of x. Select the closest answer. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ A. 17.7 B. 12.7 C. 7.6 D. 3.3 	1
3	A pair of headphones is marked at $260. In a sale it is discounted by 20%. What is the selling price? A. $312.00 B. $240.00 C. $208.00 D. $52.00	1
4	Simplify $3p^5 \times 6p^2$. A. $18p^{10}$ B. $18p^3$ C. $18p^7$ D. $9p^7$	1
5	Tom, an electrician, earns an annual salary of $68 120. How much is this per fortnight? Note: 1 year = 52 weeks = 26 fortnights = 12 months A. $1310.00 B. $2620.00 C. $2838.33 D. $5676.67	1

Short Answer:

	Question	Working Out	Mark
6	Simplify: a. $9m - 2n + m + 5n$ b. $30ab \div 5b$	a. b.	1 1
7	Expand and simplify: $5(y + 2) - 2(3y - 4)$		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
8	Factorise: a. $14m - 21$ b. $p^2 + 9p$	a. b.	1 1
9	Grace invests \$9200 for 2 years at 4.8% p.a. simple interest. Calculate the interest earned. Note: Simple interest: $I = Prn$		2
10	A jeweller buys a watch for \$250 and sells it for \$340. a. Find the profit. b. Find the profit as a percentage of the cost price. Note: Percentage profit = profit ÷ cost price × 100%	a. b.	1 1
11	Solve: $9y - 4(y + 1) = 2y + 11$		3
12	Ethan sells cars. Ethan is paid \$600 per week plus 1.5% commission on sales. Last week Ethan sold \$92 400 worth of cars. Calculate Ethan's total pay for the week.		2

	Section 3: Mixed Topics (Chapters 1–8) (continued)		
	Worded Problems:		
	Question	**Working Out**	**Mark**
13	Chloe works in a warehouse. The normal rate of pay is \$31.20 per hour for the first 38 hours each week, and time-and-a-half for each hour after that. Last week Chloe worked 46 hours.		
a	Calculate the pay for the first 38 hours.		1
b	Calculate the total pay for the 46 hours.		2
14	Bike hire costs a \$12 booking fee plus \$8 per hour. The total cost, C dollars, for h hours is given by the formula $\boldsymbol{C = 12 + 8h}$		
a	Find the total cost for 5 hours.		1
b	Mia paid \$100 altogether. Write an equation and solve it to find how many hours Mia hired the bike.		2

Section 3: Mixed Topics (Chapters 1–8) (continued)			
	Question	**Working Out**	**Mark**
15	A painter leans a ladder 4.6 m long against a vertical wall. The foot of the ladder is 1.4 m from the wall, on level ground. Note: Pythagoras' theorem: $c^2 = a^2 + b^2$ Note: $\sin\theta = \frac{\text{opp}}{\text{hyp}}$, $\cos\theta = \frac{\text{adj}}{\text{hyp}}$, $\tan\theta = \frac{\text{opp}}{\text{adj}}$ 		
a	Find h, the height the ladder reaches up the wall, correct to 2 decimal places.		2
b	Find the angle θ that the ladder makes with the ground, correct to the nearest degree.		2
	Total		**/30**

Answers

Mark your own work. If an answer is wrong, read the help sheet topic in the last column, then try the question again.

Section 1 answers – Data			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	B (The number of text messages sent in a day)	1	D1 Types of data
2	C (Symmetrical, clustered at 4 to 6)	1	D2 Shape of a distribution
3	C (3)	1	D3 Reading graphs
4	A (The mean)	1	D5 Outliers
5	C (35%)	1	D3 Reading graphs
6a	29 minutes	2	D4 Mean, median, mode, range
6b	27 minutes	1	D4 Mean, median, mode, range
7a	Jordan: 30.5; Casey: 33	2	D6 Comparing two data sets
7b	Casey: the median for Casey (33) is higher than the median for Jordan (30.5).	1	D6 Comparing two data sets
7c	Jordan: the range is 14 (37 – 23), much smaller than the range for Casey, 37 (49 – 12).	3	D6 Comparing two data sets
	Total	**14**	

Section 2 answers – Surface area and volume			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (2.35 kL)	1	M1 Units
2	C ($56\ cm^2$)	1	M2 Area of shapes
3	A ($180\ m^2$)	1	M2 Area of shapes
4	D ($156\ cm^2$)	1	M2 Area of shapes
5a	59.7 m	1	M3 Circles
5b	$283.5\ m^2$	1	M3 Circles
6a	$1272.3\ cm^3$	2	M5 Volume
6b	$692.7\ cm^2$	2	M6 Surface area
7a	76 m	1	M4 Composite shapes
7b	$252\ m^2$	3	M4 Composite shapes
7c	$16 254.00	1	M4 Composite shapes
8a	$37\ 800\ cm^3$	1	M5 Volume
8b	37.8 L	1	M1 Units
8c	$6990\ cm^2$	3	M6 Surface area
	Total	**20**	

Section 3 answers – Mixed Topics (Chapters 1–8)			
Q	**Answer**	**Marks**	**If you got it wrong, read…**
1	C (74)	1	A1 Substitution
2	B (12.7)	1	T1 Pythagoras
3	C ($208.00)	1	F1 Percentages and profit
4	C ($18p^7$)	1	A5 Index laws
5	B ($2620.00)	1	F3 Earning money
6a	$10m + 3n$	1	A2 Simplifying
6b	$6a$	1	A2 Simplifying
7	$-y + 18$	2	A3 Expanding
8a	$7(2m - 3)$	1	A4 Factorising
8b	$p(p + 9)$	1	A4 Factorising
9	$883.20	2	F2 Simple interest
10a	$90	1	F1 Percentages and profit
10b	36%	1	F1 Percentages and profit
11	$y = 5$	3	E1 Solving equations
12	$1986	2	F3 Earning money
13a	$1185.60	1	F3 Earning money
13b	$1560.00	2	F3 Earning money
14a	$52	1	E2 Formulas and word problems
14b	11 hours	2	E2 Formulas and word problems
15a	4.38 m	2	T1 Pythagoras
15b	72°	2	T2 Trigonometry
	Total	**30**	